Yoona Li

Philadelphia, PA | (215) 869-5600 | yoonali@engineering.upenn.edu

EDUCATION

University of Pennsylvania, Penn Engineering — Philadelphia, PA
Master of Applied Science in Computer Science (MAS-CS) — *Expected May 2028*

New York University — New York, NY
Bachelor of Economics & Statistics — *May 2026*

TECHNICAL SKILLS

Programming: Python, Java, C, SQL, R, MATLAB
Frameworks & Libraries: FastAPI, React, PyTorch, scikit-learn, XGBoost, NumPy, Pandas
Machine Learning: Random Forest, classification, fine-tuning, feature engineering, SHAP/LIME, model evaluation, latency/accuracy benchmarking
LLM / AI Systems: RAG, vector retrieval, streaming chat, prompt engineering, LLM evaluation, LoRA/LoRA+/QLoRA, retrieval-aware generation/abstention, LiveKit/TTS
Developer & Data Tools: Git, Docker, GitHub Actions (CI/CD); data pipelines, stateful conversational systems, reproducible testing, annotation/QC

EXPERIENCE

Lumi AI — Philadelphia, PA
Founder & Technical Product Lead — *Jun 2026 - Present*
- Adapted and extended a team-developed React/FastAPI AI tutoring platform into a teacher-grounded learning product, connecting file ingestion, vector retrieval, streamed chat, bot configuration, and real-time voice to child-facing practice and parent-facing evidence.
- Implemented retrieval-aware response boundaries and abstention behavior to reduce unsupported answers; structured retrieval, generation, safety, and evaluation as modular components for reproducible iteration across configurations.
- Built Test & Compare workflows across retrieval, prompt, and response variants, evaluating groundedness, uncertainty calibration, age-appropriate behavior, evidence of understanding, and response quality rather than relying on single prompt-level checks.
- Translated educator and preschool feedback into technical requirements for pilot workflows, product iteration, and reliability constraints, connecting software decisions to user needs and commercialization hypotheses.

ByteDance (TikTok) — Beijing, China
AI Product Manager Intern - Search Ads — *Jun 2026 - Aug 2026*
- Designed and evaluated query-similarity routing and dynamic-threshold strategies across recall, ad load, and auction quality, contributing more than $145K in incremental revenue while preserving organic-search relevance.
- Prototyped an LLM-powered Q&A advertising experience and defined evaluation logic linking answer-seeking intent, query relevance, generated-content quality, conversion, and overall search quality.

ThunderSoft — Beijing, China
Algorithm Intern — *Jun 2025 - Jul 2025*
- Built a 100K+ sample annotation/QC pipeline for a Xiaomi mobile-AI project, defining taxonomy, quality-control rules, annotator guidance, and reproducible checks to improve dataset consistency.
- Fine-tuned PyTorch models and ran reproducible latency/accuracy benchmarks, improving accuracy by approximately 8%, reducing inference latency by approximately 5%, and increasing evaluation efficiency by approximately 20%.

PROJECTS & PUBLICATION

FashionVote
Founder & Research Lead — *Aug 2025 - Dec 2025*
- Built and deployed an AI-driven fashion analytics platform using DeepFashion imagery and a crowdsourced pairwise-voting interface; curated and deduplicated image pairs and trained Random Forest models achieving 90%+ accuracy in trend classification.
- Analyzed preference data from 10K+ users, diagnosed sampling bias and label sparsity, and narrowed model interpretation toward defensible user-level recommendations; translated findings into retail product and marketing discussions.

Unified Parameter-Efficient Adaptation Framework for Quantized LLMs
First Author, IJNSEI — *2025*
- Developed a unified parameter-efficient fine-tuning framework integrating LoRA, LoRA+, and QLoRA for 4-bit quantized decoder-only LLMs, enabling billion-parameter adaptation with constrained compute.
- Built reproducible PyTorch benchmarks across trainable parameters, memory footprint, and task performance; demonstrated comparable adaptation quality with approximately 1% trainable parameters.

RESEARCH EXPERIENCE

The Wharton School — Philadelphia, PA
Research Assistant, Prof. Jonah Berger — *Jun 2026 - Aug 2026*
- Operationalized curiosity and information-gap theory into a reproducible behavioral-data annotation framework with a calibrated 7-point schema, explicit decision rules, edge-case taxonomy, and adjudication logic.
- Reviewed ambiguous and borderline cases and documented resolution criteria to improve consistency and reproducibility of downstream analysis.

NYU School of Global Public Health — New York, NY
Biostatistics Research Assistant — *Sep 2025 - May 2026*
- Designed child/family feedback surveys and a small-N ($N < 200$) tracking framework for pediatric psychological evaluations, translating clinical questions into structured binary, ordinal, categorical, and qualitative variables.
- Structured data for interpretable comparisons of length of stay by diagnosis and evaluation type while explicitly accounting for small-sample limits on inference.

NYU School of Global Public Health — New York, NY
Research Assistant, Public Health Backlash Lab — *Mar 2025 - May 2026*
- Collaborated with a cross-institutional NYU/Ohio State/University of Minnesota team; conducted Dedoose coding and analytic memoing and synthesized mixed-methods evidence on communication, policy response, community reaction, and public trust.

ENTREPRENEURSHIP & LEADERSHIP

HaoXue AI Learning Center
Founder — *May 2023 - Present*
- Founded and scaled an offline AI-powered learning and enrichment center serving 500+ students; led program design, student growth, parent engagement, and day-to-day operations.

UNICEF at NYU — New York, NY
Founder & President — *Mar 2025 - May 2026*
- Founded NYU's first UNICEF student chapter, led a 20-person team, and delivered programming including a 150+ attendee flagship event; managed outreach, partnerships, speaker coordination, and execution.